DrugRePurposeIR: Feasibility Aware Bidirectional Retrieval System

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Abstract

In this project, we will build a drug repurposing retrieval model that will help scientists find the already FDA approved drugs that can potentially be repurposed or used to treat a different neurological or infectious disease. This model will be bidirectional i.e. the query could move in two directions i.e. from a disease to candidate drug and also from a drug to candidate diseases. To cater to the bi-direction have kept the pipeline modular, so retrievers and rankers could be swapped and compared on the same queries. Results are sorted into a ranked drugs and diseases and with the feasibility aware re-ranker then shows the biologically plausible drug candidates starting with for eq: Blood Brain Barrier permeability for neurological targets and we will introduce other features that would cater to the targets. The system is scoped to neurological and infectious diseases and evaluated against known repurposing cases with a time-split test.

Introduction

Drug repurposing can reduce the time and cost associated with developing new treatments by exploring new uses for existing drugs that already have established safety information. However, the evidence of Drug Repurposing i.e. figuring out which drug has a similar causation or inhibits a certain receptor are buried in the millions of research papers across the world so a scientist needs a retrieval system that can help get down to a candidate drug (already FDA approved) with the help of targets, mechanisms and disease not just a simple keyword matching.

The motivation comes from a published study “In-silico Analysis of HIV drugs(Abacavier) for the Alzheimer's patients” whose baseline was drug repurposing: where a study suggested that workflow for alzheimer is similar as HIV so the drug given to HIV patients can be repurposed for an Alzheimer’s patient as well. In the whole study, the process of choosing a drug was manual i.e. since later is a brain disease, the drug needs to pass BBB and then with docking and simulations understand the protein-ligand interactions and if the drug could bind and stay at the target site.

That workflow was manual and drug-by-drug. This project turns its logic into a retrieval system: find literature evidence linking a drug and a disease, then filter and rank by feasibility checks like BBB permeability. Abacavir for Alzheimer's serves as the anchor case study.

Why does it matter? Approved drugs already have known human safety profiles, so a repurposing hypothesis that holds up can reach patients faster and at lower cost than a new compound. Neurological diseases such as Alzheimer's have few effective treatments and add a

hard delivery problem, the blood-brain barrier, so screening for feasibility early saves effort. Infectious diseases add urgency, because repurposing offers a quick route when a new pathogen or resistance appears.

Who benefits? Translational researchers and computational biologists who start from a disease and need candidate drugs; pharmacologists and clinicians who want to know what else a familiar drug might treat; and, indirectly, patients with few treatment options. For the course, the project also gives a concrete, measurable setting for comparing retrieval and ranking methods.

IR Task Definition

Given a drug or a disease as a query, retrieve and rank the diseases or drugs that are most plausibly connected by the literature and prefer candidates that are feasible to act on. Scientists tend to start from a disease so the system has to work both ways from one index.

Task: Ad hoc entity retrieval over a collection of scientific literature, in both directions using a single index.

Direction	Input	Output	Evidence
Drug to Disease	A drug name (good to have mechanism terms)	Ranked list of disease	Top supporting extract from each disease
Disease to Drug	A disease name (good to have pathway terms)	Ranked list of drugs	Top supporting abstract from each drug

Example Queries:

- Drug to Disease: We search abacavir, the related top ranked diseases will appear
- Disease to Drug: We search Alzheimer's, the related drugs with the BBB permeability check i.e. which drug likely to cross BBB with re-ranker comes as an output.
- Mechanism Query: We search Reverse Transcriptase inhibition, both the drug and diseases surfaced with the mechanism is displayed

Example Document:

The retrievable unit is a PubMed abstract (title and abstract text, plus its MeSH terms and normalized mentions of drugs and diseases). For example, a relevant document for the first query might be an abstract saying that an HIV reverse transcriptase inhibitor decreases neuroinflammation or amyloid-related markers in an Alzheimer’s model. An illustrative non-relevant document would be one that mentions the drug and the disease in an unrelated context such as a cohort description. These are descriptions of document types, not actual citations. Then, scores are accumulated at the paper level to get a rank for the disease or the drug.

Data

For this project, we will be using datasets that provide information about biomedical research, approved drugs, and existing drug repurposing cases. First, we will focus on neurological and infectious diseases and possibly expand to other disease areas later on. We plan to start with around 20–30 diseases, including Alzheimer's, Parkinson's, ALS, multiple sclerosis, and epilepsy for neurological diseases, and HIV, hepatitis B and C, influenza, tuberculosis, and malaria for infectious diseases. By focusing on these two categories, we will be able to explore potential drug repurposing opportunities across different diseases. This includes cases where drugs originally used for infectious diseases may also be useful for neurological conditions.

Our main source of data will be PubMed abstracts, which we will filter using Medical Subject Headings (MeSH) related to neurological and infectious diseases. These abstracts will serve as the documents used in our retrieval system. We will use DrugBank or openFDA to gather information about approved drugs. For neurological diseases, we will use the B3DB dataset to get information about whether drugs can cross the blood-brain barrier (BBB). This will help us rerank retrieved drugs based on how feasible they are for treating neurological conditions. We will also use identifiers from RxNorm or DrugBank for drugs and MeSH or MONDO for diseases to keep names consistent across datasets.

Component	Source	Use
Literature	PubMed abstracts, filtered by MeSH (Nervous System Diseases, Infections)	Documents for retrieval
Drug information	DrugBank or openFDA	Approved-drug list, targets, mechanisms
BBB labels	B3DB dataset	Feasibility feature for reranking
Drug repurposing cases	RepoDB	Ground truth, including failed attempts
Entity IDs	RxNorm or DrugBank for drugs; MeSH or MONDO for diseases	Normalizing queries and results

To determine relevance, we will use RepoDB which has information about both successful and unsuccessful drug repurposing attempts. We will determine relevance based on drug-disease pairs. For example, a pair could consist of abacavir and Alzheimer's disease. Pairs listed as

approved in RepoDB will be considered relevant if the drug was already approved for another condition. We will also include a few manually selected cases as case studies. Pairs listed as terminated or suspended will be treated as non-relevant for our evaluation. Any pairs without a known outcome will remain unjudged, since there may be potential drug repurposing opportunities that have not been studied or confirmed yet. We also plan to use a time-split evaluation, where we only retrieve information from papers published before a certain year and test whether the system can identify drug-disease relationships that were confirmed after. We will primarily use Recall@k to evaluate how well the system retrieves known relevant pairs since many pairs may not have known relevance labels. These labels will be used for evaluation purposes but an unsuccessful trial does not necessarily mean that the drug has no potential for treating the disease. We will also report the proportion of unjudged pairs in the top-k results to account for potential repurposing opportunities that have not yet been confirmed.

To prepare the data, we will first download PubMed abstracts and filter them based on the selected diseases using MeSH terms. We will then clean the text, remove duplicates and standardize drug and disease names so that information from different datasets can be matched consistently. We will use RepoDB to create our evaluation set of known drug-disease pairs. If we can't access DrugBank, we will use openFDA as an alternative source for available drug information.

Related Work

Related work is categorized into four lines: repurposing systems using knowledge graphs and language models, literature-based discovery, feasibility-aware ranking for brain diseases, and comparisons of biomedical retrievers. None of the works surveyed consider the repurposing as ranked retrieval with a controlled comparison between the retriever and the ranker families.

1. **DrugReX (2025 preprint).** builds a knowledge graph from the literature and predicts drug-disease links by link prediction with drug and disease similarity scores, with LLM-generated explanations. It is the nearest system to this project and thus acts as a baseline and source of known repurposing cases to assess.
    a. Citation: Huang, L. C., et al. "DrugReX: An Explainable Drug Repurposing System Powered by Large Language Models and Literature-Based Knowledge Graph." Research Square, 16 June 2025, https://doi.org/10.21203/rs.3.rs-6728958/v1.
2. **Deep Learning-Based Drug Repurposing Using Knowledge Graph Embeddings and GraphRAG (bioRxiv, 2025).** Utilizes pretrained TransE embeddings on the Drug Repurposing Knowledge Graph (DRKG) that contains more than 97,000 entities and 4.4 million relationships to retrieve candidate drugs for natural-language disease queries, and an LLM to explain them. It demonstrates that graph-based retrieval is feasible but not competitive with text retrievers.

    a. Citation: George, Herbert, Gowtham Baratam, and D. S. Dhyaneesh. "Deep Learning-Based Drug Repurposing Using Knowledge Graph Embeddings and GraphRAG." bioRxiv, 2025, https://doi.org/10.64898/2025.12.08.693009.
3. **KEDRec-LM (arXiv 2502.20350).** Pulls drug-disease pairs from DRKG and runs a retrieval-augmented pipeline over PubMed and clinical trial content to give each pair literature context for LLM reasoning. It combines text and structured evidence but emphasizes explanations and does not compare retrievers.
    a. Citation: Zhang, Kai, et al. "KEDRec-LM: A Knowledge-Distilled Explainable Drug Recommendation Large Language Model." arXiv, 2025, https://arxiv.org/abs/2502.20350.
4. **Literature-based discovery: addressing the issue of the subpar evaluation methodology (PMC).** Explains evaluation of literature-based discovery systems. Finding known discoveries gives real cases but few, time-sliced evaluation gives many cases but most are noise. This motivates us to combine a curated set of known pairs with a time split test, and to discuss the noise openly.
    a. Citation: Moreau, E. "Literature-Based Discovery: Addressing the Issue of the Subpar Evaluation Methodology." Bioinformatics, vol. 39, no. 2, 2023, article btad090, https://doi.org/10.1093/bioinformatics/btad090.
5. **Literature data-based de novo candidates for drug repurposing (BMC Bioinformatics, 2025).** Builds a validation set of true positives and true negatives from repoDB. Compares literature-based drug similarity with interactome-based measures . Finds literature-based Jaccard similarity performed best on AUC and F1 . It supports the use of repoDB for ground truth including its negatives.
    a. Citation: Liang, X., X. Jiang, and Y. Ma. "Literature Data-Based De Novo Candidates for Drug Repurposing." BMC Bioinformatics, vol. 26, 2025, article 203, https://doi.org/10.1186/s12859-025-06237-7.

The project tackles query-driven bidirectional repurposing search, a systematic comparison of retriever and ranker families, and a feasibility reranker evaluated on known repurposing pairs which is the gap not covered. This is claimed to our knowledge, on the basis of a non-exhaustive search.

Evaluation + Results

The evaluation of the system is on how well it ranks known repurposing pairs highly, and whether feasibility reranking helps in cases where biology suggests that it should.

Actual data: Known repurposing pairs from RepoDB by neurological and infectious diseases and hand-curated cases with abacavir as anchor for Alzheimer’s disease. Sanity checks are the classic cases: sildenafil, thalidomide, metformin(drugs). Negatives are listed for failed pairs where available.

Time Split Test: We index papers before a cutoff year, and test whether repurposings confirmed after the cutoff are in the top-k measurements

Metrics. Recall@k, MRR, and nDCG for ranking quality. For CNS queries, also report the share of non-BBB-permeable drugs in the top-k.

System	Description
Random	Randomly ranks candidate drugs or diseases to provide a baseline for comparison
BM25	Lexical baseline over abstracts, keyword matching to retrieve relevant PubMed abstracts
Dense	Biomedical Encoder + PubMed-trained retriever
Hybrid	BM25 + dense with reciprocal rank fusion
Hybrid + cross-encoder	Hybrid 1st stage, Cross encoder 2nd Stage
ABC baseline	Literature based discovery or scoring over extracted drug or disease
Hybrid + feasibility rerank	Hybrid + feasibility aware re-ranker

We will evaluate our system using Recall@k, Mean Reciprocal Rank (MRR), and normalized Discounted Cumulative Gain (nDCG). Recall@k measures how many known relevant drug-disease pairs appear in the top-k results. MRR measures how highly the first relevant result is ranked. nDCG evaluates the overall ranking quality by giving more weight to relevant results that appear higher in the list. We will use relevance judgments from RepoDB and our manually selected cases to calculate this. We will treat unjudged pairs as non-relevant for the purpose of calculating these metrics while also reporting the proportion of unjudged results in the top-k rankings. We will compare these metrics across the different retrieval methods to determine whether hybrid retrieval and feasibility-based reranking improve performance over the baselines.

Case Studies: Abacavir for Alzheimer's plus additional similar drug repurposing cases
Statistical Testing: Bootstrap or Randomized Testing for systems per query and result should have confidence since number of known pairs per disease will be small.
Results: No results exist yet. However they will be listed as each baseline system against each metric.

Work Plan

So far, we have defined our project idea and IR task, identified potential datasets, and reviewed related research on our topic. We have also outlined the retrieval methods and evaluation metrics we plan to use. Next, we will focus on collecting and preprocessing the data, implementing our baseline retrieval methods, and evaluating the system. We have outlined our weekly milestones below.

Week	Tasks	Deliverable
7-8	Finalize disease list, obtain datasets, filter PubMed abstracts, normalize entities	Initial dataset and corpus
9-10	Build evaluation set using RepoDB and implement random and BM25 baselines	Evaluation set and baseline code
11	Test baselines using Recall@k, MRR, and nDCG	Initial evaluation results
12	Implement dense and hybrid retrieval	Comparison of retrieval methods
13	Add BBB feasibility reranking	Updated ranking results
14	Run time-split evaluation and compare methods	Final evaluation results
15	Analyze results, write case studies, and finish report	Final report

Each gate is a checkpoint: the eval set must exist before baselines are scored, and baseline scores must exist before the reranker is judged against them. The Evaluation phase moves from a narrow focus on neurological and infectious diseases to a broader one.

Phase	Tasks	Deliverable
Data foundation	Set disease list and MeSH filters Download and subset PubMed Build entity normalization layer Curate ground-truth pairs from RepoDB + the anchor cases	Indexed corpus and evaluation set
Baselines	Build evaluation harness Evaluate baseline Implement BM25 and dense retrievers, and	Baseline results table

	hybrid retriever	
Reranker	Implement the aggregation strategies, add BBB and approval features, build the rule-based and then learned reranker, add the ABC baseline	Full system with swappable stages
Evaluation	Run time-split test, ablations, per disease and per direction analysis Write up case studies including abacavir Draft paper	Final results and paper draft

Future Scope:

Expand to all disease areas, add a knowledge graph (e.g. Hetionet or Open Targets) to provide drug-target-pathway explanations for the results, and add more feasibility signals, such as predicted toxicity.